

PROJECT : Healthcare Appointment Capacity & Queue Management Platform

Industry: Healthcare Operations

Detailed Problem Statement

Hospitals and clinics handle large numbers of patients across multiple departments and doctors every day. Without proper appointment and queue coordination, patients may experience excessive waiting times while doctors may have uneven workloads and available capacity may remain underutilized.

The organization requires a platform to manage doctor availability, appointment slots, patient bookings, cancellations, rescheduling, and real-time queues. Different appointment types may require different durations and departments may have different operating schedules.

The proposed **Healthcare Appointment Capacity & Queue Management Platform** should calculate available appointment capacity and organize patients into appropriate queues. The system should support priority rules while ensuring that appointment conflicts and overbooking are prevented.

Management should be able to analyze doctor utilization, appointment volumes, cancellation rates, waiting times, and department workload. The system should improve scheduling efficiency while maintaining a clear history of patient appointments.

Functional Modules

- **Patient Registration**

The system maintains patient demographic and contact information required for appointment management. Each patient receives a unique record used across appointments and visit history.

- **Doctor Management**

The system maintains doctor profiles, specialization, department, and consultation status. Doctor information is linked with availability and appointment scheduling.

- **Department Management**

The system maintains hospital departments and their operational schedules. Departments are used to organize doctors, appointment capacity, and queue information.

- **Availability Management**

The system maintains doctor availability using working days, consultation hours, breaks, and leave periods. Available capacity is calculated from configured schedules and existing appointments.

- **Appointment Management**

Patients or authorized staff can create appointments by selecting doctor, department, date, and available slot. The system prevents double booking and invalid appointment timings.

- **Queue Management**

The system creates queues for patients waiting for consultation. Queue position changes according to completed visits, cancellations, and configured priority rules.

- **Priority Management**

Authorized staff can classify appointments according to predefined priority categories. Priority rules influence queue ordering without allowing unauthorized manual manipulation.

- **Cancellation & Rescheduling**

The system supports appointment cancellation and rescheduling according to configured policies. Released slots become available for future bookings.

- **Appointment History**

The system maintains completed, cancelled, missed, and rescheduled appointments. Historical information supports patient visit tracking and operational analysis.

- **Capacity Optimization Engine**

The engine analyzes doctor schedules, appointment duration, existing bookings, and available slots. It prevents capacity conflicts and identifies suitable appointment opportunities.

- **Operational Dashboard**

Managers can monitor daily appointments, waiting queues, doctor utilization, cancellations, and department workload. Reports help identify operational bottlenecks.

- **Audit Management**

Appointment modifications and queue changes are recorded with user and timestamp information. This provides accountability for operational changes.

Core Business Engine

Appointment Slot & Queue Optimization Engine